



2026, growing at a CAGR of 6.7 percent from 2019 to 2026. Increase in emphasis on road safety, rapid demand for advanced technology in vehicles, and developments in interior lighting solutions are the prime factors for this growth.

The advent of laser technologies and advanced LEDs is set to change the face of headlights as we know it. However, there are some challenges yet to overcome. The overall cost of LEDs is far greater than traditional alternatives of halogen and high-intensity discharge (HID) lamps. However, the rise in use of LED technology is expected to bring down the cost. Moreover, supportive government regulations regarding the use of advanced lighting technologies are expected to increase the adoption of cutting-edge designs and technologies such as adaptive headlamps.

New trends in automotive lighting

Without a doubt, OLED for both interior and exterior lighting is the latest and most popular trend in the automotive lighting market. Interior ambient lamps play a vital role in a car's interior. Automobile manufacturers offer multi-colour atmosphere lamps and car owners can choose lamplight colour and brightness as per their choice, for a better experience.

Thanks to Apple, OLED has gained popularity over the last few

years. After the integration of OLED in iPhone X as a full-screen spontaneous light and low afterglow, while maintaining low power consumption, automotive manufacturers saw OLED as an opportunity.

OLED is nothing like other light sources. It uses organic matter's self-luminous feature that uses minimum energy and produces soft, low blue-light that is very similar to natural light. Moreover, OLED lighting has a low-heat value that is favourable for environmental protection. OLED's thin and light nature piqued the interest of vehicle manufacturers as it helps bring convenience and freedom to design and enables patterned lighting.

The use of OLED also helps to save space. Traditional LED lights and halogen bulbs demand space and do not offer flexibility; they cannot be embedded into the vehicle's structure. On the other hand, OLED panels offer such flexibility. As OLED panels save space and energy, the demand for OLED has increased for cars as well as aircraft and ships.

Thus, OLED technology has been the most demanding and actively penetrating technology in automotive lighting, healthcare, and various other industries. What's more, automotive companies such as Audi and BMW have already applied OLED to their automobiles and have now advanced to OLED based interior lighting for cars.

Future of automotive lighting

There is no doubt that the automotive sector is slowly moving towards complete autonomy. Thus, the interior and exterior lighting of a car will highly impact autonomous driving and it will become a means of communication. Automotive lighting will soon be responsible for communicating with drivers and other road users such as passengers in other cars, motorcycle riders, and even pedestrians. The exterior lighting will become instrumental for ensuring safety. People and wild animals at the edge of the road will be briefly illuminated by automobile lighting systems and car systems will detect the same using infrared technology.

While our eyes take care of avoiding obstacles and recognising pedestrian crossings, the case is not the same for autonomous driving. Thus, light signals in front of vehicles will help pedestrians understand if the car has recognised them and if they are free to cross the road. Although such technology is still at an early stage, it has highlighted how important exterior lighting is to an automotive.

Apart from this, Volkswagen has been experimenting with a projection of red warning lines onto the road when a car is reversing, to help pedestrians identify danger zones. This tail-light cluster is expected to bring revolution to taillights and overall exterior lighting panels as it could help in dangerous situations such as the end of a traffic jam and initialise car-to-car communication.

However, the major challenge for the future of automotive lighting is the cost and energy efficiency of light. Market players are keen to boost the reach of electric vehicles through battery technology. As the demand for electric vehicles has increased, the demand for energy-efficient lighting systems is bound to increase in the coming years. **EFY**

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